exposure to the toxic information swirling in Twitter turned the AI into a raging racist.[63]

It gets worse. In order to learn, baby algorithms need one more thing besides access to data. They also need a goal. A human baby learns how to walk because she wants to get somewhere. A lion cub learns to hunt because he wants to eat. Algorithms too must be given a goal in order to learn. In chess, it is easy to define the goal: take the opponent's king. The AI learns that sacrificing a queen for a pawn is a 'mistake', because it usually prevents the algorithm from reaching its goal. In face recognition, the goal is also easy: identify the person's gender, age and name as listed in the original database. If the algorithm guessed that George W. Bush is female, but the database says male, the goal has not been reached, and the algorithm learns from its mistake.

But if you want to train an algorithm for hiring personnel, for example, how would you define the goal? How would the algorithm know that it made a mistake and hired the 'wrong' person? We might tell the baby algorithm that its goal is to hire people who stay in the company for at least a year. Employers obviously don't want to invest a lot of time and money in training a worker who quits or gets fired after a few months. Having defined the goal in such a way, it is time to go over the data. In chess, the algorithm can produce any amount of new data just by playing against itself. But in the job market, that's impossible. Nobody can create an entire imaginary world where the baby algorithm can hire and fire imaginary people and learn from that experience. The baby algorithm can train only on an existing database about real-life people. Just as lion cubs learn what a zebra is mainly by spotting patterns in the real-life savannah, so baby algorithms learn what a good employee is by spotting patterns in real-life companies.

Unfortunately, if real-life companies already suffer from some ingrained bias, the baby algorithm is likely to learn this bias, and even amplify it. For instance, an algorithm looking for patterns of 'good employees' in real-life data may conclude that hiring the boss's nephews is always a good idea, no matter what other qualification they have. For the data clearly indicates that 'boss's nephews' are usually hired when applying for a job, and are rarely fired. The baby algorithm would spot this pattern and become nepotistic. If it is put in charge of an HR department, it will start giving preference to the boss's nephews.